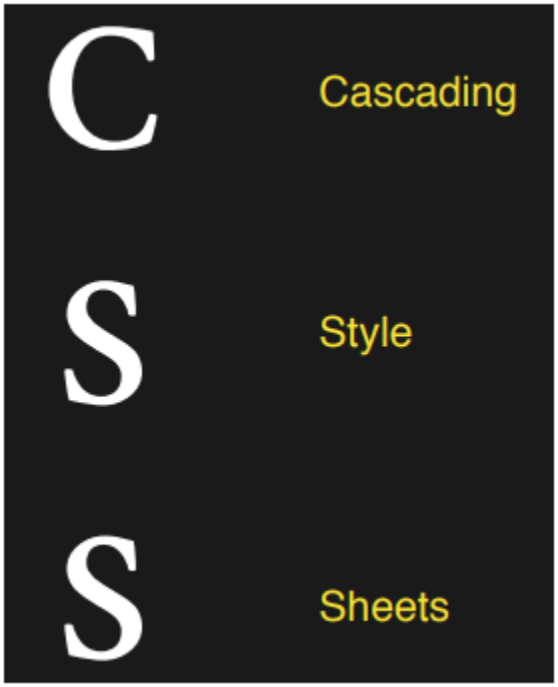
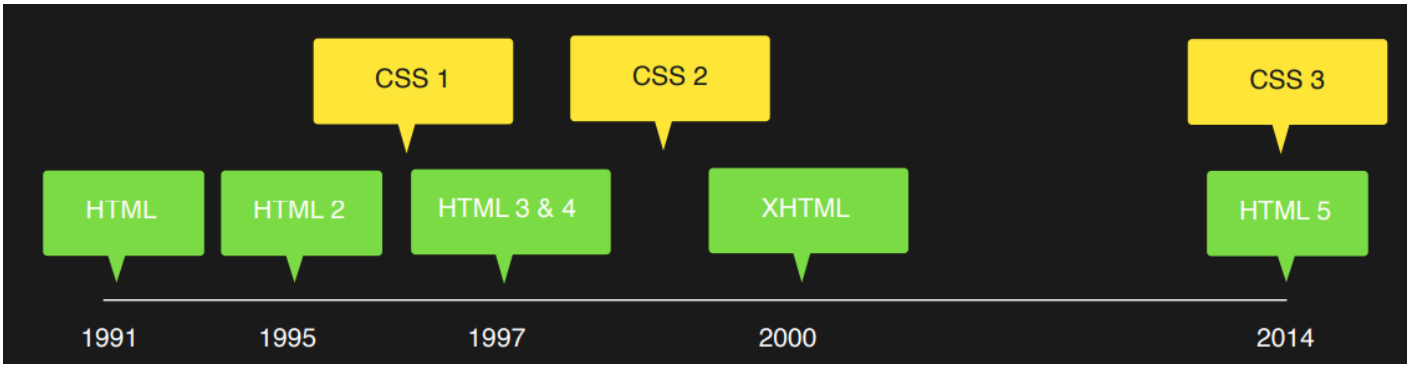
S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Web Development





HIRA KUMAR

❖ What is CSS ?

- Cascading Style Sheets (CSS) is used to format the layout of a webpage.
- CSS is generally used with HTML to change the style of web pages and user interfaces
- CSS (Cascading Style Sheets) was proposed by **Hakon Wium Lie** in 1994.
- He was working with CERN when he introduced the idea.
- The first official version, CSS1, was released in 1996 by W3C.
- With CSS, you can control the color, font, the size of text, the spacing between elements, how elements are positioned and laid out, what background images or background colors are to be used, different displays for different devices and screen sizes, and much more!
- CSS code inside an HTML page usually starts with the <style> tag. This tag is written inside the <head> section of the HTML document.
- The <style> tag tells the browser that CSS styling rules and written here.
- Inside it, We write selectors and properties to design the webpage such as can control the color, font, the size of text, the spacing between elements, how elements are positioned and laid out, what background images or background colors are to be used, different displays for different devices and screen sizes, and much more!
- When the browser reads the <style> tag, it applies those sytles to the HTML elements.
- **Syntax →**

```
<style>
        selector
                {
                        property: value;
                }
```

</style>

- **Example →**

```html
<head>
    <meta charset="UTF-8">
    <title>Document</title>
    <style>
        p{
            color: red;
            font-size: 50px;
        }
    </style>
</head>
<body>
    <p> Lorem ipsum dolor sit amet consectetur adipisicing elit. Fugit saepe libero labore laborum
    magni inventore, nobis reprehenderit quisquam amet cumque perspiciatis exercitationem
    consequatur voluptatum ipsum maiores fugiat, omnis natus! Dolorem.</p>
</body>
</html>
```

Lorem ipsum dolor sit amet consectetur adipisicing elit. Fugit saepe libero labore laborum magni inventore, nobis reprehenderit quisquam amet cumque perspiciatis exercitationem consequatur voluptatum ipsum maiores fugiat, omnis natus! Dolorem.

Note →

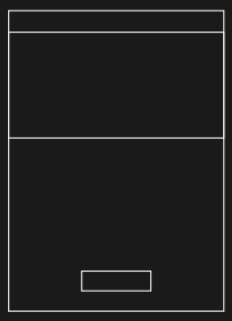
- **Sometimes the <style> tag is also used outside the <head> section in an HTML document. It can be written inside the <body> section if we want to apply styles at that place.**
- **However, it is not recommended because the standard practice is to keep CSS inside the <head> section.**
- **Writing CSS in the head helps the browser load styles before the page content appears. This keeps the code organized and easier to manage.**

What can CSS do?

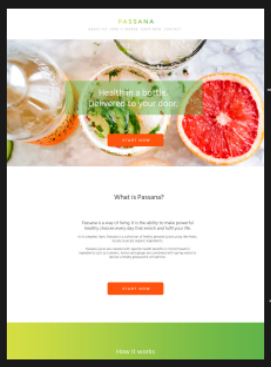
HTML only

HTML + CSS

The front end development process



Sketch / wireframe

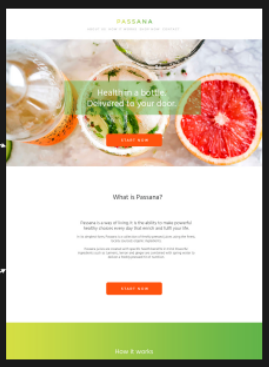


Design
(Sketch / Photoshop)

HTML
Structure

CSS
Styles

HTML / CSS
development



Design in browser

HIRA KUMAR

❖ CSS Syntax

- CSS syntax is the rule/format used to write CSS code that styles HTML elements.
- It tells the browser which element to style and what style to apply.
- **Syntax →**

```
selector
{
        property: value;
}
```

- **Example →**

```
p
{
    color: red;
}
```

Here, **p** → Selector (selects all paragraph elements)
color → Property (what you want to change)
red → Value (the color applied)

❖ Comment In CSS

✓ **Single line comment → /*……..*/**
✓ **Multi-line comment → /*…….**
……..
……..*/

Note → HTML comment not work in CSS.

```
pro > # hira.css > ...

*{
    margin: 0px;
    padding: 0px;

}
.class1{
/* single line comment*/
    background-color: rgb(141, 14, 14);
font-size : 8px;
height: 50px;
padding-top : 3px;
padding-left : 5px;
}
/* multi-line comment
.a1{
    color:white;
    text-decoration :none;
}*/
.s{
    background-color: green;
    color: white;
    text-decoration:none;
```

❖ Types of CSS based on structure

Based on Structure, CSS divided into 3 parts.

1. **Selector**
2. **Property**
3. **Value**

❖ Types of CSS

based on location/ applying method

1. **Inline CSS**

2. **Internal CSS**
3. **External CSS**

1. Inline CSS

- Inline CSS entails the application of styles directly to individual HTML elements via the use of the "style" attribute.
- **Usage:** You add the style attribute to an HTML element and define CSS rules as attribute values. These styles are applied only to the specific element.
- **Scope:** Inline CSS styles are limited to the element on which they are applied.
- **Syntax-** <tagname style="property:value";>
  Content
  </tagname>

  OR

  <tagname style="property:value; property:value….";>
  Content
  </tagname>
- **Example:-**

```html
<head>
<title> inline css</title>
</head>
<body>
<div>
    <p style="color: red"; >
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad
      asperiores molestiae facilis error alias distinctio ipsam veritatis
      eaque architecto a reiciendis ratione inventore odio, minus sunt
      aliquid maiores.
    </p>
    <p style="background-color: yellow; color: black; ">
      Hello, I am hira
      Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique
      delectus ipsa commodi nemo quidem animi, ut nostrum atque fugiat
      minima ex distinctio maxime molestiae odio sint at voluptatibus
      repudiandae neque?
    </p>
```

inline CSS

inline CSS

http://127.0.0.1:5500/index.html

Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione inventore odio, minus sunt aliquid maiores.

Hello, I am hira Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique delectus ipsa commodi nemo quidem animi, ut nostrum atque fugiat minima ex distinctio maxime molestiae odio sint at voluptatibus repudiandae neque?

➢ Advantage of inline CSS

✓ Provides granular control over individual elements' styles.
✓ Useful for quick, one-off styling for specific elements.
✓ Overrides external and internal CSS styles if needed.

Disadvantage of inline CSS

✓ Not practical for maintaining consistent styles across an entire website.
✓ Leads to code repetition if the same styles are applied to multiple elements.
✓ Can make HTML markup less clean and harder to read.

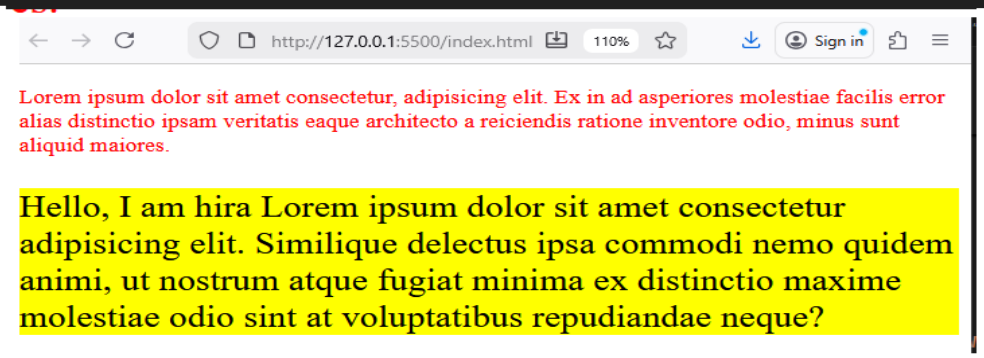
2. Internal CSS

- Internal CSS, also referred to as **embedded CSS** or document-level CSS, encompasses the practice of specifying CSS styles directly within the HTML document.
- This method typically involves placing the CSS definitions within the HTML file, often within the <head> section by employing a <style> element.

- **Embedded Cascading/css** → Internal CSS is also known as embedded cascading style sheet because Internal CSS used for multiple elements within the same HTML document.
- Syntax –

```
<html>
    <head>
        <style>
            selector{
                property:value;
            }
        </style>
    </head>
<body>
    Content of webpage
</body>
</html>
```

```html
<style>
  p{
    color: red;
  }
.hira{
  color: black;
  background-color: yellow;
  font-size:25px;
}
</style>
</head>
<body>
<div>
    <p>
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae
      facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione
      inventore odio, minus sunt aliquid maiores.
    </p>
    <p class="hira">
      Hello, I am hira
      Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique delectus ipsa commodi
      quidem animi, ut nostrum atque fugiat minima ex distinctio maxime molestiae odio sint at
      voluptatibus repudiandae neque?
    </p>
```



➢ Advantage of internal CSS

- ✓ Keeps CSS styles within the same HTML document, making it easy to maintain.
- ✓ Allows for reusable styles applied to multiple elements in the document.
- ✓ Useful for small to medium-sized projects where a single document contains all styles.

➢ Disadvantage of internal CSS

- ✓ Not as organized as external CSS for larger projects.
- ✓ Styles are not reusable across multiple HTML documents.

- ✓ Mixing content and styles in the same document can become cluttered in complex projects.

3. External CSS

- External CSS encompasses the creation of a distinct CSS file and the establishment of a connection with an HTML document.
- External CSS contains separate CSS files that contain only style properties
- Syntax –

```
<head>
    <link rel="stylesheet" href="style.css">
</head>
```

Here rel→ relationship & href → Hyper reference

➢ **Following step is a detailed set of instructions along with explanations for employing external CSS:**

Step 1: Create Your CSS File any source / make self (hira.css)
Step2 : Define CSS Styles in the External File.
Step 3: Save Your CSS File
Step 4: Create Your HTML File
Step 5: Save Your HTML File
Step 6: Place the CSS File in the Same Directory
Step 7: Link the CSS File to Your HTML Document

Example –

```css
p{
    color: red;
  }
.hira{
  color: black;
  background-color: yellow;
  font-size:25px;
}
```

External css

```html
<!DOCTYPE html>
<html>
<head>
<title> inline css</title>
<link rel="stylesheet" href="hira.css">
</head>
<body>
<div>
    <p>
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione inventore odio, minus sunt aliquid maiores.
    </p>
    <p class="hira">
      Hello, I am hira
      Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique delectus ipsa commodi nemo quidem animi, ut nostrum atque fugiat minima ex distinctio maxime molestiae odio sint at voluptatibus repudiandae neque?
    </p>
```

http://127.0.0.1:5500/index.html

Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione inventore odio, minus sunt aliquid maiores.

Hello, I am hira Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique delectus ipsa commodi nemo quidem animi, ut nostrum atque fugiat minima ex distinctio maxime molestiae odio sint at voluptatibus repudiandae neque?

❖ Priority of CSS

Inline > internal > External

- Inline → Highest Priority
- Internal → Medium
- External → low

❖ Types of CSS based on structure

1. Selectors

- Selectors are patterns used to select and target HTML elements that you want to style. CSS rules are applied to elements based on the selectors. Here are some common types of selectors –

(a) Element Selector

✓ Selects all instances of a particular HTML element.

✓ Example –

```
p
  {
        Color: red;
  }
```

(इसका मतलब style HTML में मौजूद सभी paragraphs पर अप्लाई होगा)

```
<head>
<title>hira css</title>
<style>
  p{
    color: red;
  }
</style>
</head>
<body>
<div>
    <p>
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae
      facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione
      inventore odio, minus sunt aliquid maiores.
    </p>
    <p>
      Hello, I am hira
      Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique delectus ipsa commodi nemo
      quidem animi, ut nostrum atque fugiat minima ex distinctio maxime molestiae odio sint at
      voluptatibus repudiandae neque?
    </p>
```

Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione inventore odio, minus sunt aliquid maiores.

Hello, I am hira Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique delectus ipsa commodi nemo quidem animi, ut nostrum atque fugiat minima ex distinctio maxime molestiae odio sint at voluptatibus repudiandae neque?

(b) Class Selector

✓ Selects elements with a specific class attribute. Class selectors start with a dot (.).

✓ Example

.highlight (any name can you put)

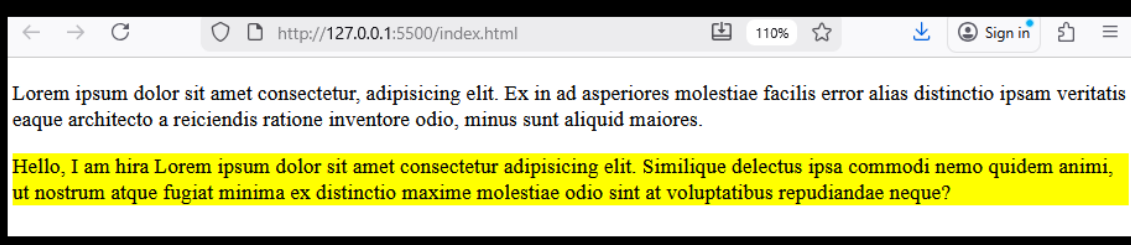
{

class="highlight"

}

(इसका मतलब जिस नाम से आप class बनाये हैं, वह सिर्फ HTML के उसी element पर अप्लाई होगा जिस element में आप use लिखेंगे या call करेंगे)

```
<head>
<style>
  .hira{
    color: black;
    background-color: yellow;
  }
</style>
</head>
<body>
<div>
    <p>
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae
      facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione
      inventore odio, minus sunt aliquid maiores.
    </p>
    <p class="hira">
      Hello, I am hira
      Lorem ipsum dolor sit amet consectetur adipisicing elit. Similique delectus ipsa commodi nemo
      quidem animi, ut nostrum atque fugiat minima ex distinctio maxime molestiae odio sint at
      voluptatibus repudiandae neque?
    </p>
```



(c) ID Selector

✓ Selects an element with a specific id attribute value. ID selectors start with a hash (#).

✓ Example

(i) #header

{

Color:blue;

}

(ii) #hira

{

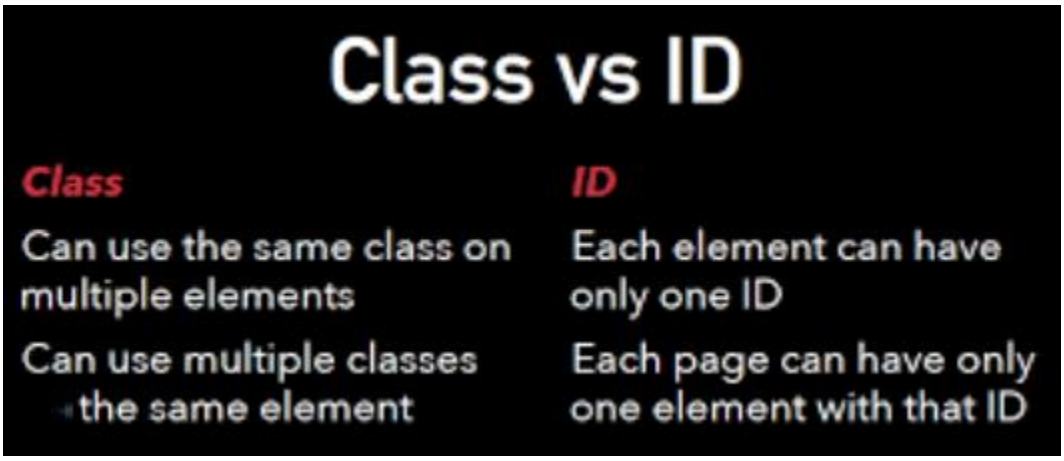
color:red;

}

(यहाँ id का use HTML के किसी unique element के लिए किया जाता है |)

Right ✅	Wrong ❌
<header id="menu"></header> <nav id="content"></nav> 	hello <a id="text"> link </a>

```
<head>
<style>
  #header{
    color: white;
    background-color: rgb(41, 120, 238);
    font-size:30px;
    text-align:center;
  }
</style>
</head>
<body>
  <div id="header">
   <a>Patliputra University,Patna</a><br>
    <a> पाटलिपुत्र यूनिवर्सिटी,पटना</a>
  </div>
<div>
    <p >
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores
      facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis
      inventore odio, minus sunt aliquid maiores.
    </p>
```

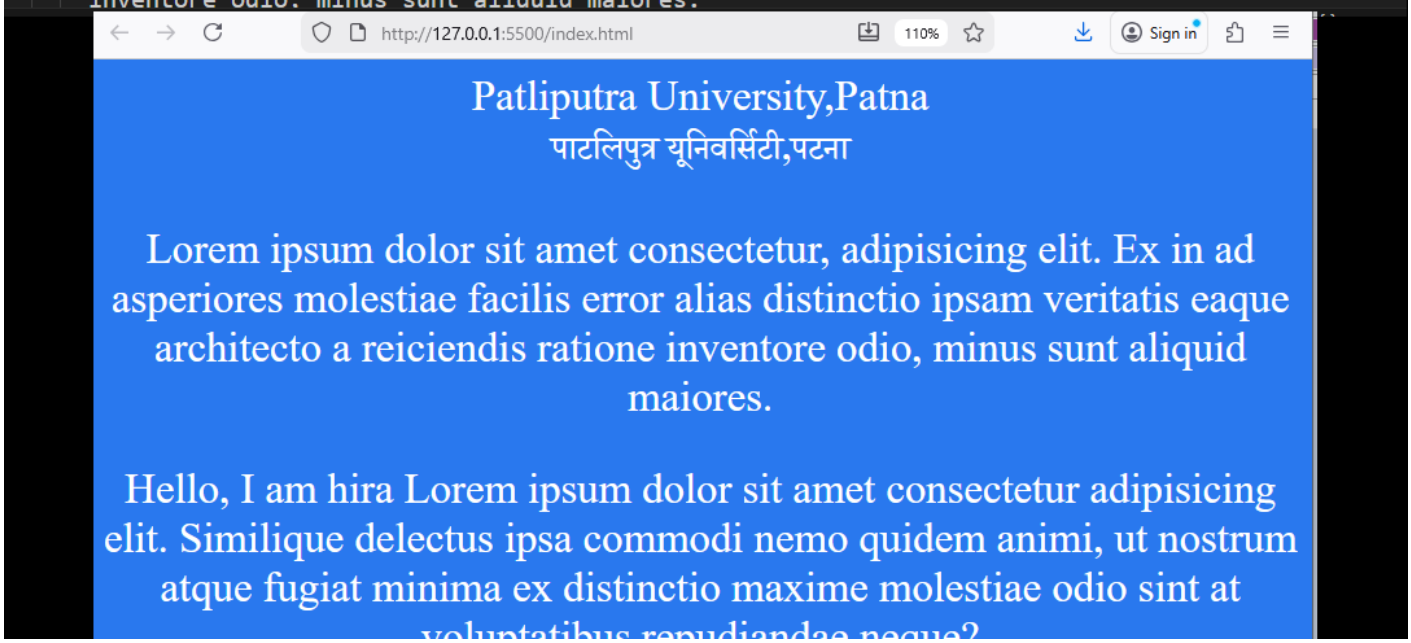


(d) Universal Selector

- ✓ The universal selector in CSS is denoted by an asterisk (*).
- ✓ When you use the universal selector, it selects every element on the page, allowing you to apply styles to all elements simultaneously.
- ✓ Example

```
*{
    Margin:0;
    }
```

```html
<title>hira css</title>
<style>
  *{
    color: white;
    background-color: rgb(41, 120, 238);
    font-size:30px;
    text-align:center;
  }
</style>
</head>
<body>
  <div >
   <a>Patliputra University,Patna</a><br>
    <a> पाटलिपुत्र यूनिवर्सिटी,पटना</a>
  </div>
<div>
    <p >
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ex in ad asperiores molestiae
      facilis error alias distinctio ipsam veritatis eaque architecto a reiciendis ratione
      inventore odio, minus sunt aliquid maiores.
```



(e) Group Selector

- Used for Select more than one element in HTML.
- Example

```
H1, h2, h3 {
        Font-weight: bold;
    }
```

```html
<style>
  p,a,div{
    color: white;
    background-color: rgb(41, 120, 238);
    font-size:30px;
    text-align:center;
  }
</style>
</head>
<body>
  <div >
   <a>Patliputra University,Patna</a><br>
    <a> पाटलिपुत्र यूनिवर्सिटी,पटना</a>
  </div>
<div>
    <p >
      HTML = Hyper Text Mark up language
    </p>
    <a >
      Hello, I am hira & CSS = Cascading style sheet
    </a>
```

http://127.0.0.1:5500/index.html 110% Sign in

Patliputra University,Patna
पाटलिपुत्र यूनिवर्सिटी,पटना

HTML = Hyper Text Mark up language

Hello, I am hira & CSS = Cascading style sheet

(f) Descendant Selector

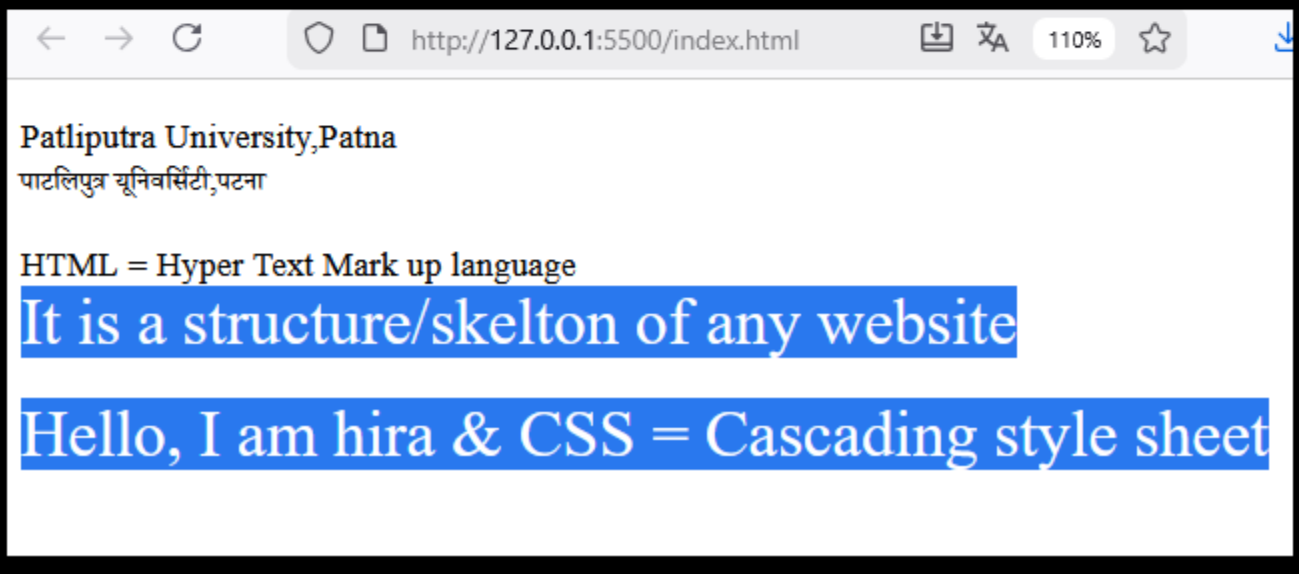
- A descendant selector in CSS is used to select all elements that are inside another element, no matter how deeply nested they are.
- It is written using a space between two selectors.
- Syntax

```
Parent element  child element {
        Property: value;
        }
```

Example

```
div p {                          OR        article > span {
        Color: red;                                font-weight: bold;
    }                                                }
```

```html
<style>
 div a{
    background-color: rgb(41, 120, 238);
    font-size:30px;
    text-align:center;
  }
</style>
</head>
<body>
  <p>
   <a>Patliputra University,Patna</a><br>
    <a> पाटलिपुत्र यूनिवर्सिटी,पटना</a>
  </p>
<div>
    <p >
      HTML = Hyper Text Mark up language <br>
      <a> It is a structure/skelton of any website</a>
    </p>
    <a >
      Hello, I am hira & CSS = Cascading style sheet
    </a>
```



(यहाँ div के a के अंदर सारा element blue होगा, div parent एवं a child हैं)

(g) Attribute selector

- An attribute selector in CSS is used to select HTML elements based on their attributes or attribute values.
- Syntax

```
selector[attribute] {
        property: value;
}
```

Note :→ space not allowed between selector & attributes

(i) Select element with a specific attribute

```
input[type] {
        border: 2px solid red;
}
```

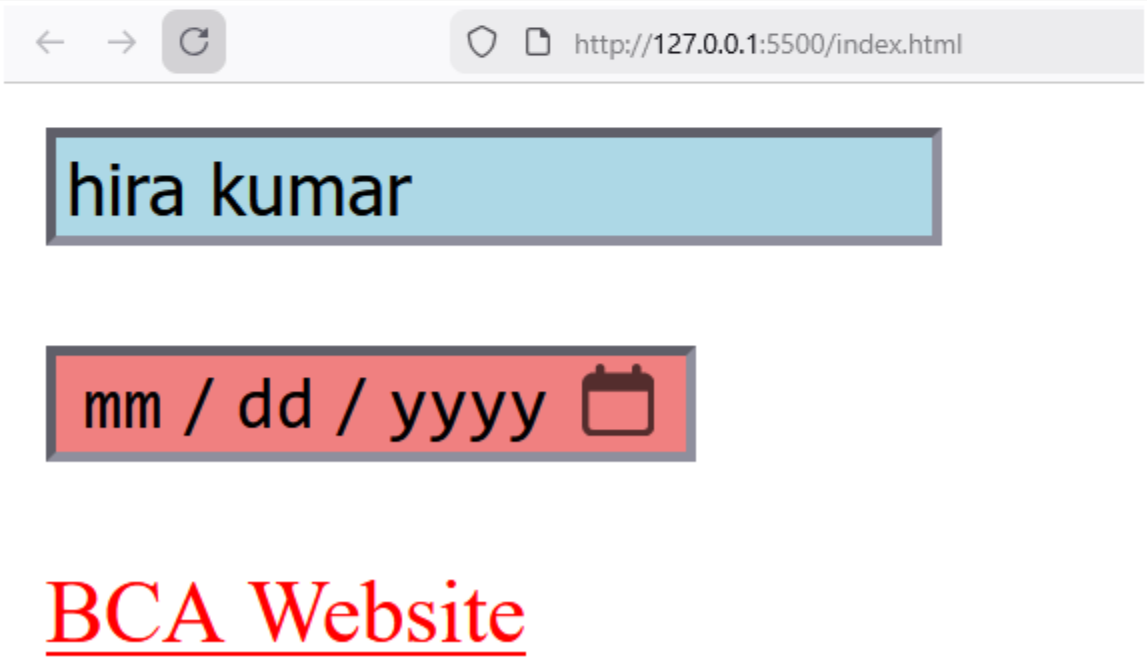
(ii) Select element with exact attribute value

```
input[type="text"] {
        background-color: yellow;
}
```

(iii) Select attribute starting with value (^)

```
a[href^="https"] {
        color: green;
}
```

```html
  <style>
    input[type]
    {
      background-color: lightblue;
    }
    input[type="date"]
        {
          background-color: lightcoral;
        }
        a[href^="https://hira24y7.github.io/bcanotes/"]
        {
          color: red;
        }
  </style>
</head>
<body>
<input type="text"> <br><br>
<input type="date"> <br><br>
<a href="https://hira24y7.github.io/bcanotes/">BCA Website</a>
</body>
```



❖ Priority of CSS in structure

1. Inline > ID > Class > Element

2. If p { color: blue; }
   p { color: red; }
   then Last color will be applied i.e. Red.
3. “Last + Specific + Important = Final Style”